Task 4: Resolving Model Hallucinations via Structured Prompting

Objective

To develop a reliable resume evaluation system using structured prompting techniques to eliminate hallucinations and ensure accurate, constraint-based output.

Libraries Used

- google-genai – for Gemini API interaction
- python-dotenv – to manage environment variables
- json – for handling structured output
- os – for file handling

Tools

- VS Code
- Python
- Gemini API

Steps Performed

1. Loaded API key securely using dotenv
2. Accepted user input (text or file-based resume)
3. Designed a structured prompt with strict rules
4. Implemented few-shot examples to guide the model
5. Added anti-prompts to prevent hallucination
6. Enforced strict JSON output format
7. Sent request to Gemini API
8. Parsed response into structured JSON
9. Displayed final output to user

Errors Faced During Execution

- Hallucination – model assumed skills not present in resume
- Extra text output – model added explanations outside JSON
- Invalid JSON format – response parsing failed
- API quota error (429) – request limit exceeded
- File not found – incorrect file path

Solutions Implemented

- Used strict prompt rules to avoid assumptions
- Added few-shot examples for better guidance
- Included anti-prompts to restrict unwanted output
- Forced JSON-only response format
- Implemented fallback mechanism when API fails
- Added error handling for invalid JSON and file issues

Output

```
PS D:\Daily Task> & "d:/Daily Task/venv/Scripts/python.exe" "d:/Daily Task/Task 4/resume_task.py"
Enter 1 for text input or 2 for file input: 1
Enter your resume text:
Lubna C++

Output:

{
    "name": "Lubna",
    "skills": [
        "C++"
    ],
    "eligibility": "Eligible"
}
```

```
PS D:\Daily Task> & "d:/Daily Task/venv/Scripts/python.exe" "d:/Daily Task/Task 4/resume_task.py"
Enter 1 for text input or 2 for file input: 2
Enter file name (example: resume.txt): resume.pdf

Output:

{
    "name": "Juien Mujawar",
    "skills": [
        "C++",
        "Python",
        "MYSQL",
        "HTML",
        "CSS",
        "JavaScript"
    ],
    "eligibility": "Eligible"
}
```

Conclusion

The system successfully reduces hallucinations using structured prompting techniques such as few-shot learning and strict constraints. It ensures reliable, consistent, and accurate resume evaluation in JSON format.